

```
<h2 id="Import-Dependencies">Import Dependencies</h2>

In [11]:
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

In [12]:
# Load dataset:
df = pd.read_csv('ppc_campaign_performance_data.csv')
df.head()

Out[12]:
<style scoped> dataframe tbody tr {<only>-of-type (vertical-align: middle); dataframe tbody tr th { (vertical-align: top); dataframe thead th { (text-align: right); </style> <thead> <th></th> <th>Campaign_ID</th> <th>Budget</th> <th>Clicks</th> <th>CTR</th> <th>CPC</th> <th>Conversions</th> <th>CPA</th> <th>Conversion_Rate</th> <th>Duration</th> <th>Revenue</th> <th>Spend</th> <th>ROAS</th> <th>Impressions</th> </thead> <tbody> <tr><td>C3876</td> <td>6300</td> <td>401</td> <td>0.0461</td> <td>15.94</td> <td>174</td> <td>36.72</td> <td>0.4539</td> <td>20</td> <td>Instagram</td> <td>Carousel</td> <td>35-44</td> <td>Male</td> <td></td> <td>Europe</td> <td>27840</td> <td>6453.9</td> <td>4.31</td> <td>2025-01-19</td> <td>8698</td>
C9725</td> <td>7700</td> <td>1684</td> <td>0.2122</td> <td>4.57</td> <td>1060</td> <td>7.26</td> <td>0.6285</td> <td>15</td> <td>Instagram</td> <td>Video</td> <td>55+</td> <td>Male</td> <td>Africa</td> <td>27840</td> <td>10067.4</td> <td>12.72</td> <td>2025-01-22</td> <td>4496</td>
C9472</td> <td>8420</td> <td>444</td> <td>0.0861</td> <td>18.96</td> <td>308</td> <td>27.34</td> <td>0.8937</td> <td>25</td> <td>Google</td> <td>Text</td> <td>25-34</td> <td>Male</td> <td>North America</td> <td>24024</td> <td>8504.2</td> <td>2.82</td> <td>2024-04-20</td> <td>4620</td>
C7601</td> <td>8470</td> <td>1912</td> <td>0.3652</td> <td>4.43</td> <td>1428</td> <td>5.93</td> <td>0.7469</td> <td>9</td> <td>Google</td> <td>Text</td> <td>25-34</td> <td>Other</td> <td>Europe</td> <td>277032</td> <td>8046.5</td> <td>34.43</td> <td>2024-08-07</td> <td>5235</td>

In [13]:
df.info()

class PandasDataFrame's
RangeIndex: 1000 entries, 0 to 999
Data columns (total 19 columns):
#   Column            Non-Null Count  Dtype
---  --
0   Campaign_ID       non-null        str
1   Budget            non-null        int64
2   Clicks            non-null        int64
3   CTR               non-null        float64
4   CPC               non-null        float64
5   Conversions       non-null        int64
6   CPA               non-null        float64
7   Conversion_Rate    non-null        float64
8   Duration          non-null        int64
9   Platform          non-null        str
10  Content_Type       non-null        str
11  Target_Age         non-null        str
12  Target_Gender      non-null        str
13  Region            non-null        str
14  Revenue           non-null        int64
15  Spend             non-null        float64
16  ROAS              non-null        float64
17  Date              non-null        str
18  Impressions       non-null        int64
dtype: float64(18), int64(6), str(17)
memory usage: 118.4 KB

In [14]:
df.isnull().sum()

Out[14]:
Campaign_ID      0
Budget           0
Clicks           0
CTR              0
CPC              0
Conversions      0
CPA              0
Conversion_Rate  0
Duration         0
Platform         0
Content_Type     0
Target_Age       0
Target_Gender    0
Region           0
Revenue          0
Spend            0
ROAS             0
Date             0
Impressions      0
dtype: int64

In [15]:
df.describe()

Out[15]:
<style scoped> dataframe tbody tr {<only>-of-type (vertical-align: middle); dataframe tbody tr th { (vertical-align: top); dataframe thead th { (text-align: right); </style> <thead> <th></th> <th>Budget</th> <th>Clicks</th> <th>CTR</th> <th>CPC</th> <th>Conversions</th> <th>CPA</th> <th>Conversion_Rate</th> <th>Duration</th> <th>Revenue</th> <th>Spend</th> <th>ROAS</th> <th>Impressions</th> </thead> <tbody> <tr><td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> </tbody> </table>

2361.091753    563.971223    0.262449    16.026811    424.76226    76.623075    0.284814    6.952796    60237.650065    2330.082996    11.1689180    2576.010494
4030.000000    50.000000    0.006100    1.080000    10.00000    1.670000    0.098600    7.000000    598.000000    1928.500000    0.070000    1000.000000
4007.500000    514.250000    0.094275    3.707500    153.75000    7.157500    0.267000    12.000000    15111.250000    4031.475000    2.717500    3262.750000
5865.000000    996.500000    0.182150    6.765000    396.50000    14.246000    0.517150    18.000000    39378.000000    5943.650000    7.296000    5410.000000
7980.000000    1491.500000    0.306900    11.285000    772.25000    38.840000    0.751675    24.000000    86009.000000    7994.850000    15.552500    7719.750000
9590.000000    1996.000000    1.785100    132.310000    1894.00000    733.080000    1.000000    30.000000    320788.000000    10468.500000    93.900000    10000.000000

In [16]:
df.columns

Out[16]:
Index(['Campaign_ID', 'Budget', 'Clicks', 'CTR', 'CPC', 'Conversions', 'CPA', 'Conversion_Rate', 'Duration', 'Platform', 'Content_Type', 'Target_Age', 'Target_Gender', 'Region', 'Revenue', 'Spend', 'ROAS', 'Date', 'Impressions'],
      dtype='object')

In [17]:
# Date format- str to Date:

df['Date'] = pd.to_datetime(df['Date'],errors='coerce')

In [18]:
df.dtypes

Out[18]:
Campaign_ID      str
Budget           int64
Clicks           int64
CTR              float64
CPC              float64
Conversions      int64
CPA              float64
Conversion_Rate  float64
Duration         int64
Platform         str
Content_Type     str
Target_Age       str
Target_Gender    str
Region          str
Revenue          int64
Spend            float64
ROAS             float64
Date             datetime64[us]
Impressions      int64
dtype: object

<h2 id="Campaign-Count-by-Platform">Campaign Count by Platform</h2>

In [10]:
platform = df.groupby('Platform').size().sort_values(ascending=False)

plt.figure(figsize=(7,3))
plt.bar(platform.index,platform.values,color='#3498db',edgecolor='black')
plt.title('Campaign Count by Platform')
plt.xlabel('Platform')
plt.ylabel('Counts')
plt.tight_layout()
plt.show()

Campaign Count by Platform

Number of Campaigns
200
150
100
50
0
LinkedIn Instagram Facebook platform YouTube Google

<h2 id="Revenue-Distribution">Revenue Distribution</h2>

In [14]:
plt.figure(figsize=(7,3))
plt.hist(df['Revenue'], bins=20)
plt.title('Revenue Distribution')
plt.xlabel('Revenue')
plt.ylabel('Number of Campaigns')
plt.tight_layout()
plt.show()

Revenue Distribution

Number of Campaigns
250
200
150
100
50
0
0 50000 100000 150000 200000 250000 300000
Revenue

<h2 id="ROAS-Distribution">ROAS Distribution</h2>

In [21]:
plt.figure(figsize=(7,3))

plt.hist(df['ROAS'],bins=20)

plt.title('ROAS Distribution')
plt.xlabel('ROAS')
plt.ylabel('Number of Campaigns')
plt.tight_layout()
plt.show()

ROAS Distribution

Number of Campaigns
250
200
150
100
50
0
0 20 40 60 80
ROAS

<h2 id="Spend vs Revenue">Spend vs Revenue</h2>

In [28]:
plt.figure(figsize=(7,4))
plt.scatter(df['Spend'],df['Revenue'],alpha=0.6,edgecolor='black')
plt.title('Spend vs Revenue')
plt.xlabel('Spend')
plt.ylabel('Revenue')
plt.tight_layout()
plt.show()

Spend vs Revenue

Revenue
300000
250000
200000
150000
100000
50000
0
2000 4000 6000 8000 10000
Spend

<h2 id="Clicks-vs-Conversions">Clicks vs Conversions</h2>

In [29]:
plt.figure(figsize=(7,4))
plt.scatter(df['Clicks'],df['Conversions'],alpha=0.6,edgecolor='black')
plt.title('Clicks vs Conversions')
plt.xlabel('Clicks')
plt.ylabel('Conversions')
plt.tight_layout()
plt.show()

Clicks vs Conversions

Conversions
1750
1500
1250
1000
750
500
250
0
0 250 500 750 1000 1250 1500 1750 2000
Clicks

In [31]:
corr = df[['Spend', 'Revenue', 'Clicks', 'Conversions', 'CTR', 'CPC', 'CPA', 'ROAS']].corr()

plt.figure(figsize=(7,5))
plt.imshow(corr, cmap='coolwarm', interpolation='nearest')
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=45)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title('Correlation Heatmap')
plt.tight_layout()
plt.show()

Correlation Heatmap

Spend Revenue Clicks Conversions CTR CPC CPA ROAS

<h1 id="Statistical-Analysis">Statistical Analysis</h1>
<h2 id="Descriptive-Stats">Descriptive Stats</h2>

In [38]:
df[['Spend', 'Revenue', 'Clicks', 'Conversions', 'CTR', 'CPC', 'CPA', 'ROAS']].describe()

Out[38]:
<style scoped> dataframe tbody tr {<only>-of-type (vertical-align: middle); dataframe tbody tr th { (vertical-align: top); dataframe thead th { (text-align: right); </style> <thead> <th></th> <th>Spend</th> <th>Revenue</th> <th>Clicks</th> <th>Conversions</th> <th>CTR</th> <th>CPC</th> <th>CPA</th> <th>ROAS</th> </thead> <tbody> <tr><td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> <td>1000.000000</td> </tbody> </table>

2361.082996    563.971223    0.262449    16.026811    424.76226    76.623075    0.284814    6.952796    60237.650065    2330.082996    11.1689180
1928.500000    598.000000    50.000000    10.00000    1.00000    1.670000    0.070000
4031.475000    15111.250000    514.250000    153.75000    0.094275    7.157500    0.267000    12.000000    39378.000000    5943.650000    7.296000
5843.650000    39378.000000    996.500000    396.50000    0.182150    14.246000    0.517150    18.000000    320788.000000    10468.500000    93.900000
10468.500000    320788.000000    1996.000000    1894.00000    1.785100    733.080000    1.000000    30.000000

<h2 id="Outlier-Detection">Outlier Detection</h2>

In [39]:
# Revenue
Q1 = df['Revenue'].quantile(0.25)
Q3 = df['Revenue'].quantile(0.75)

IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
revenue_outliers = df[
    (df['Revenue'] < lower) |
    (df['Revenue'] > upper)
]

In [39]:
revenue_outliers[
    ['Campaign_ID',
     'Platform',
     'Content_Type',
     'Region',
     'Revenue',
     'ROAS']
]

Out[39]:
<style scoped> dataframe tbody tr {<only>-of-type (vertical-align: middle); dataframe tbody tr th { (vertical-align: top); dataframe thead th { (text-align: right); </style> <thead> <th></th> <th>Campaign_ID</th> <th>Budget</th> <th>Clicks</th> <th>CTR</th> <th>CPC</th> <th>Conversions</th> <th>CPA</th> <th>Conversion_Rate</th> <th>Duration</th> <th>Revenue</th> <th>Spend</th> <th>ROAS</th> <th>Impressions</th> </thead> <tbody> <tr><td>C2514</td> <td>2810</td> <td>1491</td> <td>0.1636</td> <td>1.88</td> <td>1088</td> <td>2.63</td> <td>0.7165</td> <td>30</td> <td>Facebook</td> <td>Video</td> <td>55+</td> <td>Male</td> <td>North America</td> <td>206124</td> <td>2950.5</td> <td>69.86</td> <td>2024-06-10</td> <td>9119</td>
C3600</td> <td>4120</td> <td>1076</td> <td>0.1846</td> <td>3.83</td> <td>1027</td> <td>4.01</td> <td>0.9545</td> <td>11</td> <td>Instagram</td> <td>Image</td> <td>45-54</td> <td>Male</td> <td>South America</td> <td>189914</td> <td>3914.0</td> <td>47.76</td> <td>2024-06-04</td> <td>5629</td>
C2810</td> <td>2750</td> <td>1455</td> <td>0.2338</td> <td>1.89</td> <td>1395</td> <td>1.97</td> <td>0.9588</td> <td>12</td> <td>LinkedIn</td> <td>Image</td> <td>55+</td> <td>Male</td> <td>South America</td> <td>223200</td> <td>2777.5</td> <td>80.36</td> <td>2024-03-21</td> <td>6223</td>
C3370</td> <td>2050</td> <td>777</td> <td>0.1777</td> <td>2.66</td> <td>537</td> <td>3.85</td> <td>0.6911</td> <td>26</td> <td>Facebook</td> <td>Text</td> <td>55+</td> <td>Female</td> <td>North America</td> <td>79476</td> <td>1966.5</td> <td>40.41</td> <td>2024-10-27</td> <td>4372</td>
C2887</td> <td>2070</td> <td>1904</td> <td>1.3668</td> <td>1.08</td> <td>868</td> <td>2.36</td> <td>0.4564</td> <td>9</td> <td>Instagram</td> <td>Video</td> <td>45-54</td> <td>Other</td> <td>Europe</td> <td>155551</td> <td>2111.5</td> <td>73.67</td> <td>2024-10-10</td> <td>1393</td>

C8008</td> <td>4230</td> <td>1419</td> <td>0.2083</td> <td>2.98</td> <td>1356</td> <td>3.12</td> <td>0.9556</td> <td>12</td> <td>YouTube</td> <td>Carousel</td> <td>25-34</td> <td>Other</td> <td>Asia</td> <td>173992</td> <td>4356.9</td> <td>41.08</td> <td>2024-10-07</td> <td>6812</td>
C6513</td> <td>5630</td> <td>1788</td> <td>0.1933</td> <td>3.15</td> <td>1196</td> <td>4.71</td> <td>0.8689</td> <td>16</td> <td>YouTube</td> <td>Text</td> <td>35-44</td> <td>Other</td> <td>Europe</td> <td>230204</td> <td>6686.3</td> <td>40.80</td> <td>2024-12-15</td> <td>8249</td>
C3505</td> <td>3990</td> <td>1876</td> <td>0.3705</td> <td>2.13</td> <td>1474</td> <td>2.71</td> <td>0.7856</td> <td>8</td> <td>Facebook</td> <td>Video</td> <td>45-54</td> <td>Other</td> <td>Europe</td> <td>241736</td> <td>4029.9</td> <td>59.89</td> <td>2024-07-05</td> <td>5663</td>
C2805</td> <td>3580</td> <td>1711</td> <td>0.1768</td> <td>2.09</td> <td>1351</td> <td>2.65</td> <td>0.7897</td> <td>16</td> <td>LinkedIn</td> <td>Text</td> <td>45-54</td> <td>Male</td> <td>South America</td> <td>171577</td> <td>3558.4</td> <td>48.90</td> <td>2025-01-05</td> <td>9077</td>
C2402</td> <td>3670</td> <td>1534</td> <td>0.1974</td> <td>2.39</td> <td>1258</td> <td>2.92</td> <td>0.8201</td> <td>26</td> <td>LinkedIn</td> <td>Video</td> <td>18-24</td> <td>Other</td> <td>Asia</td> <td>206312</td> <td>3523.2</td> <td>58.56</td> <td>2024-11-08</td> <td>7771</td>

<p>64 rows × 19 columns</p>

In [41]:
# Spend Outliers
Q1 = df['Spend'].quantile(0.25)
Q3 = df['Spend'].quantile(0.75)

IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
spend_outliers = df[
    (df['Spend'] < lower) |
    (df['Spend'] > upper)
]

Out[41]:
spend_outliers

In [41]:
df[['Revenue Outliers', 'len(revenue_outliers)'],
   print('ROAS Outliers', len(roas_outliers))
   print('Spend Outliers', len(spend_outliers))

Revenue Outliers: 52
ROAS Outliers: 64
Spend Outliers: 0

In [ ]:
```